

Matrices and Systems of Linear Equations (2005–2026 Archive)

Category: Linear Algebra and Algebra • Official Mock Test Series

TOTAL QUESTIONS

75

TOTAL MARKS

155 M

TEST DURATION

225 Min

PAPER PATTERN

MCQ • MSQ • NAT
(2005–20)

Section / Type	Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Choice (MCQ)	43	+1 / +2 Marks	$-1/3$ / $-2/3$ Marks	4 Options (1 Correct)
Multiple Select (MSQ)	9	+2 Marks	Nil (0)	4 Options (1+ Correct)
Numerical Answer (NAT)	23	+1 / +2 Marks	Nil (0)	Real Decimal Value

IMPORTANT INSTRUCTIONS FOR CANDIDATES

- Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
- Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty ($1/3$ of positive marks).
- Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
- Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
- Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
- Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at vaibhavgeometer.github.io.

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2026 • Q59 • ID: JAM_2026_Q59)

NAT

+2 M

Neg: Nil (0)

Q.59

Let A be a 3×3 real matrix such that given any column vector $x \in \mathbb{R}^3$, the column vector Ax is the reflection of x about the plane

$$\{(a, b, -a - b) : a, b \in \mathbb{R}\}.$$

Then the sum of the diagonal elements of A is _____ (rounded off to one decimal place).

Question 2 (JAM 2026 • Q57 • ID: JAM_2026_Q57)

NAT

+2 M

Neg: Nil (0)

Q.57

Let $a, b \in \mathbb{R}$. If the system of linear equations

$$x + 2y + 2z = 1$$

$$2x + 3y + z = 2$$

$$ax + 5y + bz = b$$

has infinitely many solutions, then $a + b =$ _____ (rounded off to one decimal place).

Question 3 (JAM 2026 • Q56 • ID: JAM_2026_Q56)

NAT

+2 M

Neg: Nil (0)

Q.56

Let $M \in M_3(\mathbb{R})$. If

$$M \begin{pmatrix} \sqrt{15} \\ \sqrt{15} \\ 0 \end{pmatrix} = \begin{pmatrix} \sqrt{3} \\ \sqrt{2} \\ -5 \end{pmatrix}, \quad M \begin{pmatrix} 0 \\ 0 \\ \sqrt{30} \end{pmatrix} = \begin{pmatrix} \sqrt{15} \\ \sqrt{10} \\ \sqrt{5} \end{pmatrix}, \quad M \begin{pmatrix} -\sqrt{15} \\ \sqrt{15} \\ 0 \end{pmatrix} = \begin{pmatrix} \sqrt{12} \\ -\sqrt{18} \\ 0 \end{pmatrix}$$

then $\det(M) =$ _____ (rounded off to one decimal place).

Question 4 (JAM 2026 • Q50 • ID: JAM_2026_Q50)

NAT

+1 M

Neg: Nil (0)

Q.50

Let P be a 5×5 real matrix with $\det(P) = 2$. Let Q be the matrix of cofactors of P . Then $\det(Q) = \underline{\hspace{2cm}}$ (rounded off to one decimal place).

Question 5 (JAM 2026 • Q49 • ID: JAM_2026_Q49)

NAT

+1 M

Neg: Nil (0)

Q.49

Let $A \in M_3(\mathbb{C})$. Suppose the column vector $v = \begin{pmatrix} \sqrt{5}i \\ 2i \\ x \end{pmatrix}$ in $\mathbb{C}^3$ belongs to the intersection of $\text{nullspace}(A)$ and $\text{rangespace}(A^T)$.
Then $|x| = \underline{\hspace{2cm}}$ (rounded off to one decimal place).

Question 6 (JAM 2026 • Q39 • ID: JAM_2026_Q39)

MSQ

+2 M

Neg: Nil (0)

Q.39	Consider matrices P of order 4×6 and Q of order 6×4 with real entries such that $PQ = 0$ and $QP = 0$. Which of the following statements is/are TRUE?
(A)	$\text{rangespace}(P) \subseteq \text{nullspace}(Q)$ and $\text{rangespace}(Q) \subseteq \text{nullspace}(P)$.
(B)	$\text{rank}(P) + \text{rank}(Q) \leq 4$.
(C)	If $\text{rangespace}(P) = \text{nullspace}(Q)$, then $\text{rank}(P) + \text{rank}(Q) = 4$.
(D)	$\text{rangespace}(Q) = \text{nullspace}(P)$.

Question 7 (JAM 2026 • Q37 • ID: JAM_2026_Q37)

MSQ

+2 M

Neg: Nil (0)

Q.37

Let $P = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix} \in M_5(\mathbb{C})$.

Which of the following statements is/are TRUE?

- (A) $\text{nullity}(P - I) \geq 2$, where I is the 5×5 identity matrix.
- (B) P has 4 distinct eigenvalues in $\mathbb{C}$.
- (C) P has 3 distinct eigenvalues in $\mathbb{R}$.
- (D) If λ is an eigenvalue of P , then there exists a positive integer n such that $\lambda^n = 1$.

Question 8 (JAM 2026 • Q25 • ID: JAM_2026_Q25)

MCQ

+2 M

Neg: -0.67

Q.25	Let P be a nonzero complex $n \times n$ matrix such that $v^* P v \geq 0$ for all column vectors v in $\mathbb{C}^n$. Which ONE of the following statements is necessarily TRUE?
(A)	All eigenvalues of P are negative real numbers.
(B)	If v_1, v_2 are column vectors in $\mathbb{C}^n$ such that $P v_1 = v_2$ and $P v_2 = v_1$, then $v_1 = v_2$.
(C)	$v^* P v \neq 0$ for all nonzero column vectors v in $\mathbb{C}^n$.
(D)	All eigenvalues of P^* are negative real numbers.

Question 9 (JAM 2026 • Q14 • ID: JAM_2026_Q14)

MCQ

+2 M

Neg: -0.67

Q.14	Let A be a 5×3 real matrix of rank 2 and b be a non-zero 5×1 real column vector. Suppose $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ and $\begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}$ are two solutions of the system of linear equations $Ax = b$. Which ONE of the following is also a solution of $Ax = b$?
(A)	$\begin{pmatrix} 3 \\ 3 \\ 3 \end{pmatrix}$
(B)	$\begin{pmatrix} 5 \\ 7 \\ 9 \end{pmatrix}$
(C)	$\begin{pmatrix} 5 \\ 6 \\ 9 \end{pmatrix}$
(D)	$\begin{pmatrix} 7 \\ 8 \\ 9 \end{pmatrix}$

Question 10 (JAM 2026 • Q8 • ID: JAM_2026_Q8)

MCQ

+1 M

Neg: -0.33

Q.8	Let A be an invertible 5×5 real matrix. Let B be the matrix obtained by interchanging the second and third columns of A and then adding 3 times the third column to the fifth column. Which ONE of the following statements is TRUE?
(A)	B^{-1} is obtained by interchanging the second and third rows of A^{-1} , and then adding 3 times the third row to the fifth row.
(B)	B^{-1} is obtained by interchanging the second and third rows of A^{-1} , and then adding -3 times the fifth row to the third row.
(C)	B^{-1} is obtained by adding -3 times the third row to the fifth row, and then interchanging the second and third rows of A^{-1} .
(D)	B^{-1} is obtained by adding 3 times the fifth row to the third row, and then interchanging the second and third rows of A^{-1} .

Question 11 (JAM 2025 • Q22 • ID: JAM_2025_Q22)

MCQ

+2 M

Neg: -0.67

Q.22

Let $M = (m_{ij})$ be a 3×3 real, invertible matrix and $\sigma \in S_3$ be the permutation defined by $\sigma(1) = 2$, $\sigma(2) = 3$ and $\sigma(3) = 1$. The matrix $M_\sigma = (n_{ij})$ is defined by $n_{ij} = m_{i\sigma(j)}$ for all $i, j \in \{1, 2, 3\}$.

Then, which one of the following is TRUE?

- (A) $\det(M) = \det(M_\sigma)$, and nullity of the matrix $M - M_\sigma$ is 0
- (B) $\det(M) = -\det(M_\sigma)$, and nullity of the matrix $M - M_\sigma$ is 1
- (C) $\det(M) = \det(M_\sigma)$, and nullity of the matrix $M - M_\sigma$ is 1
- (D) $\det(M) = -\det(M_\sigma)$, and nullity of the matrix $M - M_\sigma$ is 0

Question 12 (JAM 2025 • Q16 • ID: JAM_2025_Q16)

MCQ

+2 M

Neg: -0.67

Q.16

Let $M = \begin{pmatrix} -2 & 0 & 0 \\ 3 & 2 & 3 \\ 4 & -1 & x \end{pmatrix}$ for some real number x . Suppose that -2 and 3 are eigenvalues of M . If $M^3 \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 125 \\ 125 \end{pmatrix}$, then which one of the following is TRUE?

- (A) $x = 5$, and the matrix $M^2 + M$ is invertible
- (B) $x \neq 5$, and the matrix $M^2 + M$ is invertible
- (C) $x = 5$, and the matrix $M^2 + M$ is NOT invertible
- (D) $x \neq 5$, and the matrix $M^2 + M$ is NOT invertible

Question 13 (JAM 2025 • Q15 • ID: JAM_2025_Q15)

MCQ

+2 M

Neg: -0.67

Q.15

Let $M = \begin{pmatrix} 6 & 2 & -6 & 8 \\ 5 & 3 & -9 & 8 \\ 3 & 1 & -2 & 4 \end{pmatrix}$. Consider the system S of linear equations given by

$$\begin{aligned} 6x_1 + 2x_2 - 6x_3 + 8x_4 &= 8 \\ 5x_1 + 3x_2 - 9x_3 + 8x_4 &= 16 \\ 3x_1 + x_2 - 2x_3 + 4x_4 &= 32 \end{aligned}$$

where x_1, x_2, x_3, x_4 are unknowns.

Then, which one of the following is TRUE?

- (A) The rank of M is 3, and the system S has a solution
- (B) The rank of M is 3, and the system S does NOT have a solution
- (C) The rank of M is 2, and the system S has a solution
- (D) The rank of M is 2, and the system S does NOT have a solution

Question 14 (JAM 2025 • Q13 • ID: JAM_2025_Q13)

MCQ

+2 M

Neg: -0.67

Q.13 Let $X = \{x \in S_4 : x^3 = id\}$ and $Y = \{x \in S_4 : x^2 \neq id\}$.

If m and n denote the number of elements in X and Y , respectively, then which one of the following is TRUE?

- (A) m is even and n is even
- (B) m is odd and n is even
- (C) m is even and n is odd
- (D) m is odd and n is odd

Question 15 (JAM 2025 • Q7 • ID: JAM_2025_Q7)

MCQ

+1 M

Neg: -0.33

Q.7

Let $M = \begin{pmatrix} 2 & 0 & -1 \\ 4 & 1 & -4 \\ 2 & 0 & x \end{pmatrix}$ for some real number x .

If 0 is an eigenvalue of M , then $(M^4 + M) \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$ is equal to

(A) $\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$

(B) $\begin{pmatrix} 2 \\ 0 \\ 2 \end{pmatrix}$

(C) $\begin{pmatrix} 5 \\ 0 \\ 5 \end{pmatrix}$

(D) $\begin{pmatrix} 17 \\ 0 \\ 17 \end{pmatrix}$

Question 16 (JAM 2024 • Q50 • ID: JAM_2024_Q50)

NAT

+1 M

Neg: Nil (0)

Q.50 Let

$$M = \begin{pmatrix} 0 & 0 & 0 & 0 & -1 \\ 2 & 0 & 0 & 0 & -4 \\ 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 3 \\ 0 & 0 & 0 & 2 & 2 \end{pmatrix}.$$

If $p(x)$ is the characteristic polynomial of M , then $p(2) - 1$ equals _____

Question 17 (JAM 2024 • Q44 • ID: JAM_2024_Q44)

NAT

+1 M

Neg: Nil (0)

Q.44 Consider the 4×4 matrix

$$M = \begin{pmatrix} 0 & 1 & 2 & 3 \\ 1 & 0 & 1 & 2 \\ 2 & 1 & 0 & 1 \\ 3 & 2 & 1 & 0 \end{pmatrix}.$$

If $a_{i,j}$ denotes the $(i,j)^{\text{th}}$ entry of M^{-1} , then $a_{4,1}$ equals _____ (rounded off to two decimal places).

Question 18 (JAM 2024 • Q37 • ID: JAM_2024_Q37)

MSQ

+2 M

Neg: Nil (0)

Q.37 For a matrix M , let $\text{Rowspace}(M)$ denote the linear span of the rows of M and $\text{Colspace}(M)$ denote the linear span of the columns of M . Which of the following hold(s) for all $A, B, C \in M_{10}(\mathbb{R})$ satisfying $A = BC$?

- (A) $\text{Rowspace}(A) \subseteq \text{Rowspace}(B)$
- (B) $\text{Rowspace}(A) \subseteq \text{Rowspace}(C)$
- (C) $\text{Colspace}(A) \subseteq \text{Colspace}(B)$
- (D) $\text{Colspace}(A) \subseteq \text{Colspace}(C)$

Question 19 (JAM 2024 • Q16 • ID: JAM_2024_Q16)

MCQ

+2 M

Neg: -0.67

Q.16 Let A be a 6×5 matrix with entries in $\mathbb{R}$ and B be a 5×4 matrix with entries in $\mathbb{R}$. Consider the following two statements.

P: For all such nonzero matrices A and B , there is a nonzero matrix Z such that AZB is the 6×4 zero matrix.

Q: For all such nonzero matrices A and B , there is a nonzero matrix Y such that BYA is the 5×5 zero matrix.

Which one of the following holds?

- (A) Both P and Q are true
- (B) P is true but Q is false
- (C) P is false but Q is true
- (D) Both P and Q are false

Question 20 (JAM 2024 • Q15 • ID: JAM_2024_Q15)

MCQ

+2 M

Neg: -0.67

Q.15

Let $a = \begin{bmatrix} \frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{6}} \\ 0 \end{bmatrix}$. Consider the following two statements.

P: The matrix $I_4 - aa^T$ is invertible.

Q: The matrix $I_4 - 2aa^T$ is invertible.

Then, which one of the following holds?

- (A) P is false but Q is true
- (B) P is true but Q is false
- (C) Both P and Q are true
- (D) Both P and Q are false

Question 21 (JAM 2024 • Q9 • ID: JAM_2024_Q9)

MCQ

+1 M

Neg: -0.33

Q.9 Consider the following statements.

P: If a system of linear equations $Ax = b$ has a unique solution, where A is an $m \times n$ matrix and b is an $m \times 1$ matrix, then $m = n$.

Q: For a subspace W of a nonzero vector space V , whenever $u \in V \setminus W$ and $v \in V \setminus W$, then $u + v \in V \setminus W$.

Which one of the following holds?

- (A) Both P and Q are true
- (B) P is true but Q is false
- (C) P is false but Q is true
- (D) Both P and Q are false

Question 22 (JAM 2023 • Q58 • ID: JAM_2023_Q58)

NAT

+2 M

Neg: Nil (0)

Q. 58 The maximum number of linearly independent eigenvectors of the matrix

$$\begin{pmatrix} 1 & 1 & 0 & 0 \\ 2 & 2 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 1 & 3 \end{pmatrix}$$

is equal to _____.

Question 23 (JAM 2023 • Q56 • ID: JAM_2023_Q56)

NAT

+2 M

Neg: Nil (0)

Q. 56 Let

$$A = \begin{pmatrix} 1 & 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 & 3 \\ 1 & 1 & 4 & 4 & 4 \end{pmatrix}$$

and B be a 5×5 real matrix such that AB is the zero matrix. Then the maximum possible rank of B is equal to _____.

Question 24 (JAM 2023 • Q14 • ID: JAM_2023_Q14)

MCQ

+2 M

Neg: -0.67

Q. 14 The system of linear equations in x_1, x_2, x_3

$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & -1 & 1 \\ 2 & 3 & \alpha \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \\ \beta \end{pmatrix}$$

where $\alpha, \beta \in \mathbb{R}$, has

- (A) at least one solution for any α and β
- (B) a unique solution for any β when $\alpha \neq 1$
- (C) NO solution for any α when $\beta \neq 5$
- (D) infinitely many solutions for any α when $\beta = 5$

Question 25 (JAM 2023 • Q13 • ID: JAM_2023_Q13)

MCQ

+2 M

Neg: -0.67

Q. 13 Let

$$A = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 0 & 0 \\ -2 & 2 & 2 \end{pmatrix}$$

and $B = A^5 + A^4 + I_3$. Which of the following is NOT an eigenvalue of B ?

- (A) 1
(B) 2
(C) 49
(D) 3

Question 26 (JAM 2023 • Q2 • ID: JAM_2023_Q2)

MCQ

+1 M

Neg: -0.33

Q. 2 Let $\mathbf{v}_1, \dots, \mathbf{v}_9$ be the column vectors of a non-zero 9×9 real matrix A . Let $a_1, \dots, a_9 \in \mathbb{R}$, not all zero, be such that $\sum_{i=1}^9 a_i \mathbf{v}_i = \mathbf{0}$. Then the system $A\mathbf{x} = \sum_{i=1}^9 \mathbf{v}_i$ has

- (A) no solution
(B) a unique solution
(C) more than one but only finitely many solutions
(D) infinitely many solutions

Question 27 (JAM 2022 • Q57 • ID: JAM_2022_Q57)

NAT

+2 M

Neg: Nil (0)

Q.57

Consider the 4×4 matrix $M = \begin{pmatrix} 11 & 10 & 10 & 10 \\ 10 & 11 & 10 & 10 \\ 10 & 10 & 11 & 10 \\ 10 & 10 & 10 & 11 \end{pmatrix}$. Then the value of the determinant of M is equal to _____.

Question 28 (JAM 2022 • Q53 • ID: JAM_2022_Q53)

NAT

+2 M

Neg: Nil (0)

Q.53

Let $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \\ -1 & 1 \end{pmatrix}$ and let A^T denote the transpose of A . Let $u = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$ and $v = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}$ be column vectors with entries in $\mathbb{R}$ such that $u_1^2 + u_2^2 = 1$ and $v_1^2 + v_2^2 + v_3^2 = 1$. Suppose

$$Au = \sqrt{2} v \quad \text{and} \quad A^T v = \sqrt{2} u.$$

Then $|u_1 + 2\sqrt{2} v_1|$ is equal to _____. (Rounded off to two decimal places)

Question 29 (JAM 2022 • Q30 • ID: JAM_2022_Q30)

MCQ

+2 M

Neg: -0.67

Q.30

Let P be a 3×3 real matrix having eigenvalues $\lambda_1 = 0, \lambda_2 = 1$ and $\lambda_3 = -1$. Further, $v_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, $v_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$ and $v_3 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$ are eigenvectors of the matrix P corresponding to the eigenvalues λ_1, λ_2 and λ_3 , respectively. Then the entry in the first row and the third column of P is

(A) 0

(B) 1

(C) -1

(D) 2

Question 30 (JAM 2022 • Q26 • ID: JAM_2022_Q26)

MCQ

+2 M

Neg: -0.67

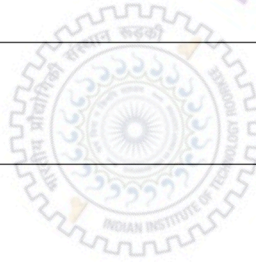
Q.26

For a 4×4 matrix $M \in M_4(\mathbb{C})$, let $\bar{M}$ denote the matrix obtained from M by replacing each entry of M by its complex conjugate. Consider the real vector space

$$H = \{M \in M_4(\mathbb{C}) : M^T = \bar{M}\}$$

where M^T denotes the transpose of M . The dimension of H as a vector space over $\mathbb{R}$ is equal to

- (A) 6
- (B) 16
- (C) 15
- (D) 12



Question 31 (JAM 2022 • Q20 • ID: JAM_2022_Q20)

MCQ

+2 M

Neg: -0.67

Q.20	For $P \in M_5(\mathbb{R})$ and $i, j \in \{1, 2, \dots, 5\}$, let p_{ij} denote the $(i, j)^{\text{th}}$ entry of P. Let $S = \{P \in M_5(\mathbb{R}) : p_{ij} = p_{rs} \text{ for } i, j, r, s \in \{1, 2, \dots, 5\} \text{ with } i + r = j + s\}.$ Then which one of the following is FALSE?
(A)	S is a subspace of the vector space over $\mathbb{R}$ of all 5×5 symmetric matrices.
(B)	The dimension of S over $\mathbb{R}$ is 5.
(C)	The dimension of S over $\mathbb{R}$ is 11.
(D)	If $P \in S$ and all the entries of P are integers, then 5 divides the sum of all the diagonal entries of P .

Question 32 (JAM 2022 • Q18 • ID: JAM_2022_Q18)

MCQ

+2 M

Neg: -0.67

Q.18

Consider the system of linear equations

$$\begin{aligned}x + y + t &= 4, \\ 2x - 4t &= 7, \\ x + y + z &= 5, \\ x - 3y - z - 10t &= \lambda,\end{aligned}$$

where x, y, z, t are variables and λ is a constant. Then which one of the following is true?

- (A) If $\lambda = 1$, then the system has a unique solution.
- (B) If $\lambda = 2$, then the system has infinitely many solutions.
- (C) If $\lambda = 1$, then the system has infinitely many solutions.
- (D) If $\lambda = 2$, then the system has a unique solution.

Question 33 (JAM 2022 • Q17 • ID: JAM_2022_Q17)

MCQ

+2 M

Neg: -0.67

Q.17	For $X, Y \in M_2(\mathbb{R})$, define $(X, Y) = XY - YX$. Let $\mathbf{0} \in M_2(\mathbb{R})$ denote the zero matrix. Consider the two statements: $P : (X, (Y, Z)) + (Y, (Z, X)) + (Z, (X, Y)) = \mathbf{0}$ for all $X, Y, Z \in M_2(\mathbb{R})$. $Q : (X, (Y, Z)) = ((X, Y), Z)$ for all $X, Y, Z \in M_2(\mathbb{R})$. Then which one of the following is correct?
(A)	Both P and Q are true.
(B)	P is true, but Q is false.
(C)	P is false, but Q is true.
(D)	Both P and Q are false.

Question 34 (JAM 2022 • Q16 • ID: JAM_2022_Q16)

MCQ

+2 M

Neg: -0.67

Q.16	Let $P \in M_4(\mathbb{R})$ be such that P^4 is the zero matrix, but P^3 is a nonzero matrix. Then which one of the following is FALSE?
(A)	For every nonzero vector $v \in \mathbb{R}^4$, the subset $\{v, Pv, P^2v, P^3v\}$ of the real vector space $\mathbb{R}^4$ is linearly independent.
(B)	The rank of P^k is $4 - k$ for every $k \in \{1, 2, 3, 4\}$.
(C)	0 is an eigenvalue of P .
(D)	If $Q \in M_4(\mathbb{R})$ is such that Q^4 is the zero matrix, but Q^3 is a nonzero matrix, then there exists a nonsingular matrix $S \in M_4(\mathbb{R})$ such that $S^{-1}QS = P$.

Question 35 (JAM 2022 • Q2 • ID: JAM_2022_Q2)

MCQ

+1 M

Neg: -0.33

Q.2	The rank of the 4×6 matrix $\begin{pmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{pmatrix}$ with entries in $\mathbb{R}$, is
(A)	1
(B)	2
(C)	3
(D)	4

Question 36 (JAM 2022 • Q1 • ID: JAM_2022_Q1)

MCQ

+1 M

Neg: -0.33

Q.1	Consider the 2×2 matrix $M = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \in M_2(\mathbb{R})$. If the eighth power of M satisfies $M^8 \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$, then the value of x is
(A)	21
(B)	22
(C)	34
(D)	35

Question 37 (JAM 2021 • Q59 • ID: JAM_2021_Q59)

NAT

+2 M

Neg: Nil (0)

Q. 59 Let $A = \begin{pmatrix} 2 & -1 & 3 \\ 2 & -1 & 3 \\ 3 & 2 & -1 \end{pmatrix}$. Then the largest eigenvalue of A is ____.

Question 38 (JAM 2021 • Q56 • ID: JAM_2021_Q56)

NAT

+2 M

Neg: Nil (0)

JAM 2021

MATHEMATICS - MA

Q. 56 The determinant of the matrix

$$\begin{pmatrix} 2021 & 2020 & 2020 & 2020 \\ 2021 & 2021 & 2020 & 2020 \\ 2021 & 2021 & 2021 & 2020 \\ 2021 & 2021 & 2021 & 2021 \end{pmatrix}$$

is ____.

Question 39 (JAM 2021 • Q40 • ID: JAM_2021_Q40)

MSQ

+2 M

Neg: Nil (0)

Q. 40 Let $m > 1$ and $n > 1$ be integers. Let A be an $m \times n$ matrix such that for some $m \times 1$ matrix b_1 , the equation $Ax = b_1$ has infinitely many solutions. Let b_2 denote an $m \times 1$ matrix different from b_1 . Then $Ax = b_2$ has

- (A) infinitely many solutions for some b_2 . (B) a unique solution for some b_2 .
 (C) no solution for some b_2 . (D) finitely many solutions for some b_2 .



MA

13 / 17

Question 40 (JAM 2021 • Q22 • ID: JAM_2021_Q22)

MCQ

+2 M

Neg: -0.67

Q. 22 Let A be an $n \times n$ invertible matrix and C be an $n \times n$ nilpotent matrix. If $X = \begin{pmatrix} X_{11} & X_{12} \\ X_{21} & X_{22} \end{pmatrix}$ is a $2n \times 2n$ matrix (each X_{ij} being $n \times n$) that commutes with the $2n \times 2n$ matrix $B = \begin{pmatrix} A & 0 \\ 0 & C \end{pmatrix}$, then

- (A) X_{11} and X_{22} are necessarily zero matrices.
 (B) X_{12} and X_{21} are necessarily zero matrices.
 (C) X_{11} and X_{21} are necessarily zero matrices.
 (D) X_{12} and X_{22} are necessarily zero matrices.

Admission test for Masters 2021
 Institute:
 of Science Bangalore

Question 41 (JAM 2021 • Q10 • ID: JAM_2021_Q10)

MCQ

+1 M

Neg: -0.33

JAM 2021

MATHEMATICS - MA

Q. 10 Let $n > 1$ be an integer. Consider the following two statements for an arbitrary $n \times n$ matrix A with complex entries.

- I. If $A^k = I_n$ for some integer $k \geq 1$, then all the eigenvalues of A are k^{th} roots of unity.
II. If, for some integer $k \geq 1$, all the eigenvalues of A are k^{th} roots of unity, then $A^k = I_n$.

Then

- (A) both I and II are TRUE. (B) I is TRUE but II is FALSE.
(C) I is FALSE but II is TRUE. (D) neither I nor II is TRUE.



MA

4 / 17

Question 42 (JAM 2020 • Q58 • ID: JAM_2020_Q58)

NAT

+2 M

Neg: Nil (0)

Q. 58 Let $M = \begin{bmatrix} 9 & 2 & 7 & 1 \\ 0 & 7 & 2 & 1 \\ 0 & 0 & 11 & 6 \\ 0 & 0 & -5 & 0 \end{bmatrix}$. Then, the value of $\det((8I - M)^3)$ is _____.

Question 43 (JAM 2020 • Q36 • ID: JAM_2020_Q36)

MSQ

+2 M

Neg: Nil (0)

Q. 36 Consider the following system of linear equations

$$x + y + 5z = 3, \quad x + 2y + mz = 5 \quad \text{and} \quad x + 2y + 4z = k.$$

The system is consistent if

(A) $m \neq 4$ (B) $k \neq 5$ (C) $m = 4$ (D) $k = 5$

MA

12 / 17

Question 44 (JAM 2020 • Q23 • ID: JAM_2020_Q23)

MCQ

+2 M

Neg: -0.67

Q. 23 Let M be a 4×3 real matrix and let $\{e_1, e_2, e_3\}$ be the standard basis of $\mathbb{R}^3$. Which of the following is true?

(A) If $\text{rank}(M) = 1$, then $\{Me_1, Me_2\}$ is a linearly independent set(B) If $\text{rank}(M) = 2$, then $\{Me_1, Me_2\}$ is a linearly independent set(C) If $\text{rank}(M) = 2$, then $\{Me_1, Me_3\}$ is a linearly independent set(D) If $\text{rank}(M) = 3$, then $\{Me_1, Me_3\}$ is a linearly independent set

Question 45 (JAM 2020 • Q20 • ID: JAM_2020_Q20)

MCQ

+2 M

Neg: -0.67

Q. 20 Let M be an $n \times n$ ($n \geq 2$) non-zero real matrix with $M^2 = 0$ and let $\alpha \in \mathbb{R} \setminus \{0\}$. Then

- (A) α is the only eigenvalue of $(M + \alpha I)$ and $(M - \alpha I)$
 (B) α is the only eigenvalue of $(M + \alpha I)$ and $(\alpha I - M)$
 (C) $-\alpha$ is the only eigenvalue of $(M + \alpha I)$ and $(M - \alpha I)$
 (D) $-\alpha$ is the only eigenvalue of $(M + \alpha I)$ and $(\alpha I - M)$

MA

7 / 17

Question 46 (JAM 2020 • Q19 • ID: JAM_2020_Q19)

MCQ

+2 M

Neg: -0.67

Q. 19 Let M be a real 6×6 matrix. Let 2 and -1 be two eigenvalues of M . If $M^5 = aI + bM$, where $a, b \in \mathbb{R}$, then

- (A) $a = 10, b = 11$ (B) $a = -11, b = 10$
 (C) $a = -10, b = 11$ (D) $a = 10, b = -11$

Question 47 (JAM 2019 • Q58 • ID: JAM_2019_Q58)

NAT

+2 M

Neg: Nil (0)

Q.58 Let M be a 3×3 matrix with real entries such that $M^2 = M + 2I$, where I denotes the 3×3 identity matrix. If α, β and γ are eigenvalues of M such that $\alpha\beta\gamma = -4$, then $\alpha + \beta + \gamma$ is equal to _____

Question 48 (JAM 2019 • Q57 • ID: JAM_2019_Q57)

NAT

+2 M

Neg: Nil (0)

Q.57 Let M and N be any two 4×4 matrices with integer entries satisfying

$$MN = 2 \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Then the maximum value of $\det(M) + \det(N)$ is _____

Question 49 (JAM 2019 • Q56 • ID: JAM_2019_Q56)

NAT

+2 M

Neg: Nil (0)

Q.56 If $\begin{pmatrix} 2 \\ y \\ z \end{pmatrix}$, $y, z \in \mathbb{R}$, is an eigenvector corresponding to a real eigenvalue of the matrix $\begin{pmatrix} 0 & 0 & 2 \\ 1 & 0 & -4 \\ 0 & 1 & 3 \end{pmatrix}$ then $z - y$ is equal to _____

Question 50 (JAM 2019 • Q43 • ID: JAM_2019_Q43)

NAT

+1 M

Neg: Nil (0)

Q.43 Consider the following system of three linear equations in four unknowns x_1, x_2, x_3 and x_4

$$\begin{aligned} x_1 + x_2 + x_3 + x_4 &= 4, \\ x_1 + 2x_2 + 3x_3 + 4x_4 &= 5, \\ x_1 + 3x_2 + 5x_3 + kx_4 &= 5. \end{aligned}$$

If the system has no solutions, then $k =$ _____

Question 51 (JAM 2019 • Q27 • ID: JAM_2019_Q27)

MCQ

+2 M

Neg: -0.67

Q.27 The set of eigenvalues of which one of the following matrices is NOT equal to the set of eigenvalues of $\begin{pmatrix} 1 & 2 \\ 4 & 3 \end{pmatrix}$?

(A) $\begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix}$

(B) $\begin{pmatrix} 3 & 2 \\ 4 & 1 \end{pmatrix}$

(C) $\begin{pmatrix} 3 & 4 \\ 2 & 1 \end{pmatrix}$

(D) $\begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix}$

Question 52 (JAM 2019 • Q12 • ID: JAM_2019_Q12)

MCQ

+2 M

Neg: -0.67

Q.12 Let P be a 4×4 matrix with entries from the set of rational numbers. If $\sqrt{2} + i$, with $i = \sqrt{-1}$, is a root of the characteristic polynomial of P and I is the 4×4 identity matrix, then

(A) $P^4 = 4P^2 + 9I$ (B) $P^4 = 4P^2 - 9I$ (C) $P^4 = 2P^2 - 9I$ (D) $P^4 = 2P^2 + 9I$

Question 53 (JAM 2018 • Q37 • ID: JAM_2018_Q37)

MSQ

+2 M

Neg: Nil (0)

Q.37 Let $m, n \in \mathbb{N}$, $m < n$, $P \in M_{n \times m}(\mathbb{R})$, $Q \in M_{m \times n}(\mathbb{R})$. Then which of the following is (are) NOT possible?

- (A) $\text{rank}(PQ) = n$
- (B) $\text{rank}(QP) = m$
- (C) $\text{rank}(PQ) = m$
- (D) $\text{rank}(QP) = \left\lceil \frac{m+n}{2} \right\rceil$, the smallest integer larger than or equal to $\frac{m+n}{2}$

Question 54 (JAM 2018 • Q36 • ID: JAM_2018_Q36)

MSQ

+2 M

Neg: Nil (0)

Q.36 Suppose $\alpha, \beta, \gamma \in \mathbb{R}$. Consider the following system of linear equations.

$x + y + z = \alpha$, $x + \beta y + z = \gamma$, $x + y + \alpha z = \beta$. If this system has at least one solution, then which of the following statements is (are) TRUE?

- (A) If $\alpha = 1$ then $\gamma = 1$
- (B) If $\beta = 1$ then $\gamma = \alpha$
- (C) If $\beta \neq 1$ then $\alpha = 1$
- (D) If $\gamma = 1$ then $\alpha = 1$

Question 55 (JAM 2018 • Q27 • ID: JAM_2018_Q27)

MCQ

+2 M

Neg: -0.67

Q.27 Let I denote the 4×4 identity matrix. If the roots of the characteristic polynomial of a 4×4 matrix

M are $\pm \sqrt{\frac{1 \pm \sqrt{5}}{2}}$, then $M^8 =$

- (A) $I + M^2$
- (B) $2I + M^2$
- (C) $2I + 3M^2$
- (D) $3I + 2M^2$

Question 56 (JAM 2017 • Q58 • ID: JAM_2017_Q58)

NAT

+2 M

Neg: Nil (0)

Q.58 Let $\alpha, \beta, \gamma, \delta$ be the eigenvalues of the matrix

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{bmatrix}$$

Then $\alpha^2 + \beta^2 + \gamma^2 + \delta^2 =$ _____

Question 57 (JAM 2017 • Q48 • ID: JAM_2017_Q48)

NAT

+1 M

Neg: Nil (0)

Q.48 Let P be a 7×7 matrix of rank 4 with real entries. Let $\mathbf{a} \in \mathbb{R}^7$ be a column vector. Then the rank of $P + \mathbf{a}\mathbf{a}^T$ is at least _____

Question 58 (JAM 2017 • Q38 • ID: JAM_2017_Q38)

MSQ

+2 M

Neg: Nil (0)

Q.38 Let M be an $n \times n$ matrix with real entries such that $M^3 = I$. Suppose that $M\mathbf{v} \neq \mathbf{v}$ for any non-zero vector $\mathbf{v}$. Then which of the following statements is / are TRUE?

- (A) M has real eigenvalues
- (B) $M + M^{-1}$ has real eigenvalues
- (C) n is divisible by 2
- (D) n is divisible by 3

Question 59 (JAM 2017 • Q34 • ID: JAM_2017_Q34)

MSQ

+2 M

Neg: Nil (0)

Q.34 If X and Y are $n \times n$ matrices with real entries, then which of the following is/are TRUE?

- (A) If $P^{-1}XP$ is diagonal for some real invertible matrix P , then there exists a basis for $\mathbb{R}^n$ consisting of eigenvectors of X
- (B) If X is diagonal with distinct diagonal entries and $XY = YX$, then Y is also diagonal
- (C) If X^2 is diagonal, then X is diagonal
- (D) If X is diagonal and $XY = YX$ for all Y , then $X = \lambda I$ for some $\lambda \in \mathbb{R}$

Question 60 (JAM 2017 • Q12 • ID: JAM_2017_Q12)

MCQ

+2 M

Neg: -0.67

Q.12 Let $\mathcal{M}$ be the set of all invertible 5×5 matrices with entries 0 and 1. For each $M \in \mathcal{M}$, let $n_1(M)$ and $n_0(M)$ denote the number of 1's and 0's in M , respectively. Then

$$\min_{M \in \mathcal{M}} |n_1(M) - n_0(M)| =$$

- (A) 1 (B) 3 (C) 5 (D) 15

Question 61 (JAM 2016 • Q53 • ID: JAM_2016_Q53)

NAT

+2 M

Neg: Nil (0)

Q.53 Let

$$M = \begin{bmatrix} \alpha & 1 & 1 \\ 1 & \beta & 1 \\ 1 & 1 & \gamma \end{bmatrix}, \quad \alpha\beta\gamma = 1, \quad \alpha, \beta, \gamma \in \mathbb{R} \quad \text{and} \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in \mathbb{R}^3.$$

Then $M\mathbf{x} = \mathbf{0}$ has infinitely many solutions if $\text{trace}(M)$ is _____

Question 62 (JAM 2016 • Q43 • ID: JAM_2016_Q43)

NAT

+1 M

Neg: Nil (0)

Q.43 Let $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in \mathbb{R}^3$ be a non-zero vector and $A = \frac{\mathbf{x} \mathbf{x}^T}{\mathbf{x}^T \mathbf{x}}$. Then the dimension of the vector space $\{\mathbf{y} \in \mathbb{R}^3 \mid A\mathbf{y} = \mathbf{0}\}$ over $\mathbb{R}$ is _____

Question 63 (JAM 2016 • Q27 • ID: JAM_2016_Q27)

MCQ

+2 M

Neg: -0.67

Q.27 Let $M = \begin{bmatrix} 0 & 1 & -2 \\ -1 & 0 & \alpha \\ 2 & -\alpha & 0 \end{bmatrix}$, $\alpha \in \mathbb{R} \setminus \{0\}$ and $\mathbf{b}$ a non-zero vector such that $M\mathbf{x} = \mathbf{b}$ for some $\mathbf{x} \in \mathbb{R}^3$. Then the value of $\mathbf{x}^T \mathbf{b}$ is

(A) $-\alpha$ (B) α (C) 0 (D) 1

Question 64 (JAM 2016 • Q26 • ID: JAM_2016_Q26)

MCQ

+2 M

Neg: -0.67

Q.26 The matrix $M = \begin{bmatrix} \cos \alpha & \sin \alpha \\ i \sin \alpha & i \cos \alpha \end{bmatrix}$ is a unitary matrix when α is

(A) $(2n+1)\frac{\pi}{2}, n \in \mathbb{Z}$ (B) $(3n+1)\frac{\pi}{3}, n \in \mathbb{Z}$

(C) $(4n+1)\frac{\pi}{4}, n \in \mathbb{Z}$ (D) $(5n+1)\frac{\pi}{5}, n \in \mathbb{Z}$

Question 65 (JAM 2016 • Q13 • ID: JAM_2016_Q13)

MCQ

+2 M

Neg: -0.67

Q.13 The largest eigenvalue of the matrix $\begin{bmatrix} 1 & 4 & 16 \\ 4 & 16 & 1 \\ 16 & 1 & 4 \end{bmatrix}$ is

(A) 16 (B) 21

(C) 48 (D) 64

Question 66 (JAM 2015 • Q60 • ID: JAM_2015_Q60)

NAT

+2 M

Neg: Nil (0)

Q.20 Let P and Q be two real matrices of size 4×6 and 5×4 , respectively. If $\text{rank}(Q) = 4$ and $\text{rank}(QP) = 2$, then $\text{rank}(P)$ is equal to _____

END OF THE QUESTION PAPER

MA

16/16

Question 67 (JAM 2015 • Q16 • ID: JAM_2015_Q16)

MCQ

+2 M

Neg: -0.67

Q.16

Let $A = \begin{bmatrix} 0 & 1-i \\ -1-i & i \end{bmatrix}$ and $B = A^T \bar{A}$. Then

- (A) an eigenvalue of B is purely imaginary
- (B) an eigenvalue of A is zero
- (C) all eigenvalues of B are real
- (D) A has a non-zero real eigenvalue

Question 68 (JAM 2015 • Q3 • ID: JAM_2015_Q3)

MCQ

+1 M

Neg: -0.33

Q.3 Let a, b, c, d be distinct non-zero real numbers with $a + b = c + d$. Then an eigenvalue of the matrix $\begin{bmatrix} a & b & 1 \\ c & d & 1 \\ 1 & -1 & 0 \end{bmatrix}$ is

- (A) $a + c$
- (B) $a + b$
- (C) $a - b$
- (D) $b - d$

Question 69 (JAM 2014 • Q12 • ID: JAM_2014_Q12)

MCQ

+2 M

Neg: -0.67

- Q.12 Let $A = \begin{bmatrix} a & -1 & 4 \\ 0 & b & 7 \\ 0 & 0 & 3 \end{bmatrix}$ be a matrix with real entries. If the sum and the product of all the eigenvalues of A are 10 and 30 respectively, then $a^2 + b^2$ equals
- (A) 29 (B) 40 (C) 58 (D) 65

Question 70 (JAM 2014 • Q11 • ID: JAM_2014_Q11)

MCQ

+2 M

Neg: -0.67

- Q.11 The system of linear equations

$$x - y + 2z = b_1$$

$$x + 2y - z = b_2$$

$$2y - 2z = b_3$$

is inconsistent when (b_1, b_2, b_3) equals

- (A) (2, 2, 0) (B) (0, 3, 2) (C) (2, 2, 1) (D) (2, -1, -2)

Question 71 (JAM 2012 • Q12 • ID: JAM_2012_Q12)

MCQ

+6 M

Neg: -2.0

- Q.12 Let P be a 4×4 matrix whose determinant is 10. The determinant of the matrix $-3P$ is

- (A) -810 (B) -30 (C) 30 (D) 810

Question 72 (JAM 2011 • Q11 • ID: JAM_2011_Q11)

MCQ

+6 M

Neg: -2.0

- Q.11 Let A be a 3×3 matrix with $\text{trace}(A) = 3$ and $\det(A) = 2$. If 1 is an eigenvalue of A , then the eigenvalues of the matrix $A^2 - 2I$ are

- (A) 1, $2(i-1)$, $-2(i+1)$ (B) -1, $2(i-1)$, $2(i+1)$
 (C) 1, $2(i+1)$, $-2(i+1)$ (D) -1, $2(i-1)$, $-2(i+1)$

Question 73 (JAM 2007 • Q11 • ID: JAM_2007_Q11)

MCQ

+6 M

Neg: -2.0

11. The rank of the matrix $\begin{bmatrix} 1 & 4 & 8 \\ 2 & 10 & 22 \\ 0 & 4 & 12 \end{bmatrix}$ is

- (A) 3
- (B) 2
- (C) 1
- (D) 0

Question 74 (JAM 2006 • Q11 • ID: JAM_2006_Q11)

MCQ

+6 M

Neg: -2.0

11. If A and B are 3×3 real matrices such that $\text{rank}(AB)=1$, then $\text{rank}(BA)$ **cannot** be

- (A) 0
- (B) 1
- (C) 2
- (D) 3

Question 75 (JAM 2005 • Q6 • ID: JAM_2005_Q6)

MCQ

+6 M

Neg: -2.0

6. Let A be a 3×3 matrix with eigenvalues 1, -1 and 3. Then

- (A) $A^2 + A$ is non-singular
- (B) $A^2 - A$ is non-singular
- (C) $A^2 + 3A$ is non-singular
- (D) $A^2 - 3A$ is non-singular

OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

Matrices and Systems of Linear Equations (2005–2026 Archive) • All Official Answers & Acceptance Ranges

Q. No.	Type	Marks	Official Key	Q. No.	Type	Marks	Official Key	Q. No.	Type	Marks	Official Key
1	NAT	+2	1.0 to 1.0	26	MCQ	+1	D	51	MCQ	+2	D
2	NAT	+2	6.0 to 6.0	27	NAT	+2	41 to 41	52	MCQ	+2	C
3	NAT	+2	−1.0 to −1.0	28	NAT	+2	3.00 to 3.00	53	MSQ	+2	A;D
4	NAT	+1	16.0 to 16.0	29	MCQ	+2	C	54	MSQ	+2	A;B
5	NAT	+1	3.0 to 3.0	30	MCQ	+2	B	55	MCQ	+2	C
6	MSQ	+2	A;B;C	31	MCQ	+2	C	56	NAT	+2	5.9 to 6.1
7	MSQ	+2	A;B;D	32	MCQ	+2	C	57	NAT	+1	2.9 to 3.1
8	MCQ	+2	B	33	MCQ	+2	B	58	MSQ	+2	B, C
9	MCQ	+2	D	34	MCQ	+2	A	59	MSQ	+2	A, B, D
10	MCQ	+1	B	35	MCQ	+1	D	60	MCQ	+2	A
11	MCQ	+2	C	36	MCQ	+1	C	61	NAT	+2	3.0 to 3.0
12	MCQ	+2	B	37	NAT	+2	4	62	NAT	+1	2.0 to 2.0
13	MCQ	+2	A	38	NAT	+2	2021	63	MCQ	+2	C
14	MCQ	+2	B	39	MSQ	+2	MTA	64	MCQ	+2	A
15	MCQ	+1	B	40	MCQ	+2	B	65	MCQ	+2	B
16	NAT	+1	31 to 31	41	MCQ	+1	B	66	NAT	+2	2
17	NAT	+1	0.15 to 0.18	42	NAT	+2	−216 to −216	67	MCQ	+2	C
18	MSQ	+2	B;C	43	MSQ	+2	A;D	68	MCQ	+1	B
19	MCQ	+2	A	44	MCQ	+2	D	69	MCQ	+2	A
20	MCQ	+2	A	45	MCQ	+2	B	70	MCQ	+2	C
21	MCQ	+1	D	46	MCQ	+2	A	71	MCQ	+6	D
22	NAT	+2	3 to 3	47	NAT	+2	3 to 3	72	MCQ	+6	D
23	NAT	+2	2 to 2	48	NAT	+2	17 to 17	73	MCQ	+6	B
24	MCQ	+2	B	49	NAT	+2	3 to 3	74	MCQ	+6	D
25	MCQ	+2	B	50	NAT	+1	7 to 7	75	MCQ	+6	C

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt (−1/3 for 1-mark questions, −2/3 for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g., $[a, b]$) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.